Parametric Survival Models

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Configuring R

Functions from these packages will be used throughout this document:

```
library(conflicted) # check for conflicting function definitions
# library(printr) # inserts help-file output into markdown output
library(rmarkdown) # Convert R Markdown documents into a variety of formats.
library(pander) # format tables for markdown
library(ggplot2) # graphics
library(ggfortify) # help with graphics
library(dplyr) # manipulate data
library(tibble) # `tibble`s extend `data.frame`s
library(magrittr) # `%>%` and other additional piping tools
library(haven) # import Stata files
library(knitr) # format R output for markdown
library(tidyr) # Tools to help to create tidy data
library(plotly) # interactive graphics
library(dobson) # datasets from Dobson and Barnett 2018
library(parameters) # format model output tables for markdown
library(haven) # import Stata files
library(latex2exp) # use LaTeX in R code (for figures and tables)
library(fs) # filesystem path manipulations
library(survival) # survival analysis
library(survminer) # survival analysis graphics
library(KMsurv) # datasets from Klein and Moeschberger
library(parameters) # format model output tables for
library(webshot2) # convert interactive content to static for pdf
library(forcats) # functions for categorical variables ("factors")
```

```
library(stringr) # functions for dealing with strings
library(lubridate) # functions for dealing with dates and times
library(broom) # Summarizes key information about statistical objects in tidy tibbles
library(broom.helpers) # Provides suite of functions to work with regression model 'broom::tidy()' t
```

Here are some R settings I use in this document:

```
rm(list = ls()) # delete any data that's already loaded into R

conflicts_prefer(dplyr::filter)
ggplot2::theme_set(
  ggplot2::theme_bw() +
        # ggplot2::labs(col = "") +
    ggplot2::theme(
      legend.position = "bottom",
      text = ggplot2::element_text(size = 12, family = "serif")))

knitr::opts_chunk$set(message = FALSE)
options('digits' = 6)

panderOptions("big.mark", ",")
pander::panderOptions("table.emphasize.rownames", FALSE)
pander::panderOptions("table.split.table", Inf)
conflicts_prefer(dplyr::filter) # use the `filter()` function from dplyr() by default
legend_text_size = 9
run_graphs = TRUE
```

1 Parametric Survival Models

Video lecture

David Rocke's Epi 204 lecture *Parametric Survival Regression*[a] (slides[b]) covers exponential and Weibull models, accelerated failure time models, the `survreg()` function, and complementary log-log plots, using the `anderson` data [1].

[a] https://dmrocke.ucdavis.edu/Class/EPI204-Spring-2021/EPI204-2021-05-20.mp4
[b] https://dmrocke.ucdavis.edu/Class/EPI204-Spring-2021/Lecture%20Parametric%20Survival%20Models.pdf

1.1 Exponential Distribution

The exponential distribution[1] is the baseline parametric model in survival analysis. It assumes a constant hazard rate[2] $\lambda > 0$ over time. This constant hazard implies that the event probability in any short interval depends only on the interval duration and not on elapsed time (the memoryless property).

[1] https://morrison-lab.github.io/pds/random-variables.html#def-exponential
[2] intro-to-survival-analysis.qmd#def-hazard

$$
\begin{aligned}
\mathrm{f}(t) &\stackrel{\text{def}}{=} \lambda \mathrm{e}^{-\lambda t} && \text{(probability density function definition for } t \geq 0) \\
\log\{\mathrm{f}(t)\} &= \log\{\lambda\} - \lambda t && \text{(taking natural logarithm of density)} \\
\mathrm{F}(t) &\stackrel{\text{def}}{=} \int_0^t \mathrm{f}(u)du && \text{(cumulative distribution function definition)} \\
&= \int_0^t \lambda \mathrm{e}^{-\lambda u} du && \text{(substituting probability density function)} \\
&= \left[-\mathrm{e}^{-\lambda u}\right]_0^t && \text{(evaluating antiderivative)} \\
&= 1 - \mathrm{e}^{-\lambda t} && \text{(evaluating limits of integration)} \\
\mathrm{S}(t) &\stackrel{\text{def}}{=} 1 - \mathrm{F}(t) && \text{(survival function definition)} \\
&= 1 - (1 - \mathrm{e}^{-\lambda t}) && \text{(substituting cumulative distribution function)} \\
&= \mathrm{e}^{-\lambda t} && \text{(simplifying terms)} \\
\Lambda(t) &\stackrel{\text{def}}{=} -\log\{\mathrm{S}(t)\} && \text{(cumulative hazard function definition)} \\
&= -\log\{\mathrm{e}^{-\lambda t}\} && \text{(substituting survival function)} \\
&= \lambda t && \text{(simplifying logarithm of exponential)} \\
\lambda(t) &\stackrel{\text{def}}{=} \frac{\partial}{\partial t}\Lambda(t) && \text{(hazard function definition as derivative of cumulative hazard)} \\
&= \frac{\partial}{\partial t}(\lambda t) && \text{(substituting cumulative hazard)} \\
&= \lambda && \text{(differentiating linear term)} \\
\mathbb{E}[T] &\stackrel{\text{def}}{=} \int_0^\infty \mathrm{S}(t)dt && \text{(expectation formula for non-negative continuous random variable)} \\
&= \int_0^\infty \mathrm{e}^{-\lambda t} dt && \text{(substituting survival function)} \\
&= \left[-\frac{1}{\lambda}\mathrm{e}^{-\lambda t}\right]_0^\infty && \text{(evaluating antiderivative)} \\
&= \lambda^{-1} && \text{(evaluating limits of integration)}
\end{aligned}
$$

Theorem 1.1 (The exponential distribution is memoryless)**.** *If T has an exponential distribution with rate $\lambda > 0$, then for all $s \geq 0$ and $t \geq 0$,*

$$\mathbb{P}(T > s + t \mid T > s) = \mathbb{P}(T > t).$$

Proof

Proof.

$$
\begin{aligned}
\mathbb{P}(T > s + t \mid T > s) &= \frac{\mathbb{P}(T > s + t,\ T > s)}{\mathbb{P}(T > s)} && \text{(definition of conditional probability)} \\
&= \frac{\mathbb{P}(T > s + t)}{\mathbb{P}(T > s)} && \text{(the event } T > s + t \text{ implies } T > s) \\
&= \frac{\mathrm{S}(s + t)}{\mathrm{S}(s)} && \text{(definition of the survival function)} \\
&= \frac{\mathrm{e}^{-\lambda(s+t)}}{\mathrm{e}^{-\lambda s}} && \text{(exponential survival function)} \\
&= \mathrm{e}^{-\lambda(s+t)+\lambda s} && \text{(quotient rule for exponentials)} \\
&= \mathrm{e}^{-\lambda t} && \text{(cancel } \lambda s) \\
&= \mathbb{P}(T > t) && \text{(exponential survival function)}
\end{aligned}
$$

□

Example 1.1 (Remaining survival after 10 event-free weeks)**.** Let T be exponential with rate $\lambda = 0.1$ events per week. For a subject who has been event-free for $s = 10$ weeks, the probability of lasting at least $t = 5$ more weeks is:

$$\begin{aligned}
\mathbb{P}(T > 15 \mid T > 10) &= \frac{\mathrm{e}^{-0.1\cdot 15}}{\mathrm{e}^{-0.1\cdot 10}} && \text{(ratio of survival functions)} \\
&= \frac{\mathrm{e}^{-1.5}}{\mathrm{e}^{-1}} && \text{(multiply)} \\
&= \mathrm{e}^{-0.5} && \text{(quotient rule for exponentials)} \\
&\approx 0.6065 && \text{(evaluate)}
\end{aligned}$$

This value equals $\mathbb{P}(T > 5) = \mathrm{e}^{-0.1\cdot 5} = \mathrm{e}^{-0.5}$, the probability that a new subject lasts 5 weeks: the 10 weeks already survived do not change the outlook.

Example 1.2 (A Weibull distribution with shape 2 is not memoryless (counterexample))**.** Now let T be Weibull with $\lambda = 0.1$ and $p = 2$, so $\mathrm{S}(t) = \mathrm{e}^{-(0.1t)^2}$ and the hazard increases over time. Then:

$$\begin{aligned}
\mathbb{P}(T > 15 \mid T > 10) &= \frac{\mathrm{e}^{-(0.1\cdot 15)^2}}{\mathrm{e}^{-(0.1\cdot 10)^2}} && \text{(ratio of survival functions)} \\
&= \frac{\mathrm{e}^{-(1.5)^2}}{\mathrm{e}^{-(1)^2}} && \text{(multiply)} \\
&= \frac{\mathrm{e}^{-2.25}}{\mathrm{e}^{-1}} && \text{(square)} \\
&= \mathrm{e}^{-1.25} && \text{(quotient rule for exponentials)} \\
&\approx 0.2865 && \text{(evaluate)}
\end{aligned}$$

whereas $\mathbb{P}(T > 5) = \mathrm{e}^{-(0.5)^2} = \mathrm{e}^{-0.25} \approx 0.7788$. A subject who has already lasted 10 weeks is less likely to last 5 more weeks than a new subject is, because the hazard has risen in the meantime.

1.2 Weibull Distribution

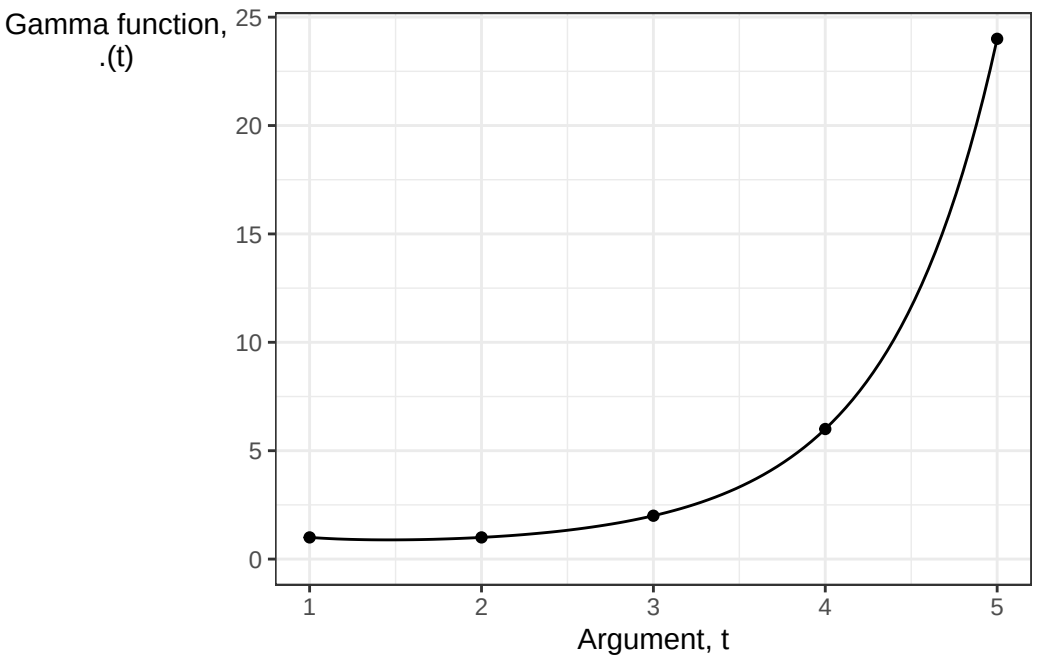
The Weibull distribution[3] generalizes the exponential distribution by introducing a shape parameter $p > 0$ alongside the scale parameter $\lambda > 0$ [2]. This flexibility allows the hazard rate to change monotonically over time, accommodating situations where risk increases or decreases as time elapses.

Using the parameterization of Kalbfleisch and Prentice [2]:

[3] https://morrison-lab.github.io/pds/distributions.html#def-weibull

$$
\begin{aligned}
\mathrm{f}(t) &\stackrel{\text{def}}{=} \lambda p(\lambda t)^{p-1}\mathrm{e}^{-(\lambda t)^p} && \text{(probability density function for } t \geq 0\text{)} \\
\mathrm{F}(t) &\stackrel{\text{def}}{=} \int_0^t \mathrm{f}(u)du && \text{(cumulative distribution function definition)} \\
&= \int_0^t \lambda p(\lambda u)^{p-1}\mathrm{e}^{-(\lambda u)^p}du && \text{(substituting density function)} \\
&= \left[-\mathrm{e}^{-(\lambda u)^p}\right]_0^t && \text{(evaluating antiderivative via substitution } w = (\lambda u)^p\text{)} \\
&= 1 - \mathrm{e}^{-(\lambda t)^p} && \text{(evaluating limits of integration)} \\
\mathrm{S}(t) &\stackrel{\text{def}}{=} 1 - \mathrm{F}(t) && \text{(survival function definition)} \\
&= 1 - \left(1 - \mathrm{e}^{-(\lambda t)^p}\right) && \text{(substituting cumulative distribution function)} \\
&= \mathrm{e}^{-(\lambda t)^p} && \text{(simplifying terms)} \\
\Lambda(t) &\stackrel{\text{def}}{=} -\log\{\mathrm{S}(t)\} && \text{(cumulative hazard function definition)} \\
&= -\log\left\{\mathrm{e}^{-(\lambda t)^p}\right\} && \text{(substituting survival function)} \\
&= (\lambda t)^p && \text{(simplifying logarithm of exponential)} \\
\log\{\Lambda(t)\} &= \log\{(\lambda t)^p\} && \text{(taking natural logarithm of cumulative hazard)} \\
&= p\log\{\lambda t\} && \text{(applying exponent rule for logarithms)} \\
&= p\log\{\lambda\} + p\log\{t\} && \text{(applying product rule for logarithms)} \\
\lambda(t) &\stackrel{\text{def}}{=} \frac{\partial}{\partial t}\Lambda(t) && \text{(hazard function definition)} \\
&= \frac{\partial}{\partial t}((\lambda t)^p) && \text{(substituting cumulative hazard)} \\
&= p(\lambda t)^{p-1} \cdot \lambda && \text{(applying power and chain rules of differentiation)} \\
&= \lambda p(\lambda t)^{p-1} && \text{(rearranging factors)} \\
\mathbb{E}[T] &\stackrel{\text{def}}{=} \int_0^\infty \mathrm{S}(t)dt && \text{(expectation formula for non-negative random variable)} \\
&= \int_0^\infty \mathrm{e}^{-(\lambda t)^p}dt && \text{(substituting survival function)} \\
&= \frac{1}{\lambda p}\int_0^\infty u^{\frac{1}{p}-1}\mathrm{e}^{-u}du && \text{(substituting } u = (\lambda t)^p \implies t = \lambda^{-1}u^{1/p}\text{)} \\
&= \lambda^{-1} \cdot \Gamma\left(1 + \frac{1}{p}\right) && \text{(applying definition of Gamma function } \Gamma(z)\text{)}
\end{aligned}
$$

> **Note**
>
> Recall from calculus:
> - $\Gamma(t) \stackrel{\text{def}}{=} \int_{u=0}^\infty u^{t-1}\mathrm{e}^{-u}du$
> - $\Gamma(t) = (t-1)!$ for integers $t \in \mathbb{Z}$
> - It is implemented by the `gamma()` function in R.



Here are some Weibull density functions[4], with $\lambda = 1$ and p varying:

```
library(ggplot2)
lambda <- 1
ggplot() +
  geom_function(
    aes(col = "0.25"),
    fun = \(x) dweibull(x, shape = 0.25, scale = 1 / lambda)
  ) +
  geom_function(
    aes(col = "0.5"),
    fun = \(x) dweibull(x, shape = 0.5, scale = 1 / lambda)
  ) +
  geom_function(
    aes(col = "1"),
    fun = \(x) dweibull(x, shape = 1, scale = 1 / lambda)
  ) +
  geom_function(
    aes(col = "1.5"),
    fun = \(x) dweibull(x, shape = 1.5, scale = 1 / lambda)
  ) +
  geom_function(
    aes(col = "2"),
    fun = \(x) dweibull(x, shape = 2, scale = 1 / lambda)
  ) +
  geom_function(
    aes(col = "5"),
    fun = \(x) dweibull(x, shape = 5, scale = 1 / lambda)
  ) +
  theme_bw() +
  xlim(0, 2.5) +
  ylab("Density,\nf(t)") +
  theme(axis.title.y = element_text(angle = 0)) +
  theme(legend.position = "bottom") +
```

[4] https://morrison-lab.github.io/pds/random-variables.html#def-pdf

```
guides(
  col =
    guide_legend(
      title = "p",
      label.theme = element_text(size = 12)
    )
)
```

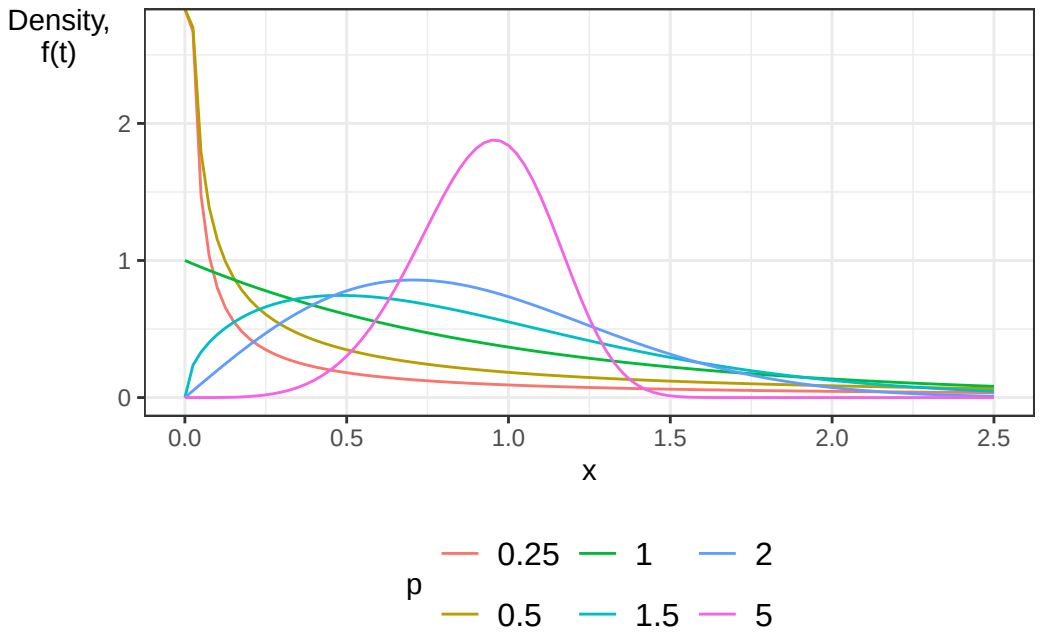


Figure 1: Density functions for Weibull distribution

1.2.1 Properties of Weibull hazard functions

Theorem 1.2 (Properties of Weibull hazard functions)**.** *If T has a Weibull distribution, then:*

- *When $p = 1$, the Weibull distribution simplifies to the exponential distribution.*
- *When $p > 1$, the hazard is strictly increasing: $\lambda'(t) > 0$.*
- *When $p < 1$, the hazard is strictly decreasing: $\lambda'(t) < 0$.*
- *$\log \Lambda(t)$ is a straight line relative to $\log t$: $\log \Lambda(t) = p \log \lambda + p \log t$.*

Proof

Proof. We prove each property of the Weibull hazard function $\lambda(t) = \lambda p(\lambda t)^{p-1}$:

1. **Simplification to Exponential distribution when $p = 1$:** Setting $p = 1$ in the Weibull hazard function yields:

$$\begin{aligned} \lambda(t) &= \lambda(1)(\lambda t)^{1-1} && \text{(substituting } p = 1 \text{ into Weibull hazard formula)} \\ &= \lambda \cdot 1 \cdot (\lambda t)^0 && \text{(simplifying exponent } 1 - 1 = 0) \\ &= \lambda && \text{(since } (\lambda t)^0 = 1 \text{ for } t > 0) \end{aligned}$$

This constant value is the hazard function[a] of the exponential distribution with parameter λ.

2. **Increasing hazard when $p > 1$**: Differentiating $\lambda(t) = \lambda^p p t^{p-1}$ with respect to t gives:

$$\begin{aligned} \lambda'(t) &\overset{\text{def}}{=} \frac{\partial}{\partial t}(\lambda^p p t^{p-1}) && \text{(definition of hazard derivative)} \\ &= \lambda^p p(p-1)t^{p-2} && \text{(applying power rule of differentiation)} \end{aligned}$$

   Since $\lambda > 0$, $p > 0$, and $t > 0$, the sign of $\lambda'(t)$ depends entirely on the factor $(p-1)$. When $p > 1$, $(p-1) > 0$, so $\lambda'(t) > 0$ for all $t > 0$, meaning the hazard function is strictly increasing over time.
3. **Decreasing hazard when $p < 1$**: Using the derivative $\lambda'(t) = \lambda^p p(p-1)t^{p-2}$: When $p < 1$, $(p-1) < 0$, so $\lambda'(t) < 0$ for all $t > 0$, meaning the hazard function is strictly decreasing over time.
4. **Linear relationship between $\log \Lambda(t)$ and $\log t$**: Taking the natural logarithm[b] of the cumulative hazard function[c] $\Lambda(t) = (\lambda t)^p$:

$$\begin{aligned} \log\{\Lambda(t)\} &\overset{\text{def}}{=} \log\{(\lambda t)^p\} && \text{(substituting Weibull cumulative hazard)} \\ &= p\log\{\lambda t\} && \text{(applying logarithm exponent rule } \log(a^b) = b\log a\text{)} \\ &= p\log\{\lambda\} + p\log\{t\} && \text{(applying logarithm product rule } \log(ab) = \log a + \log b\text{)} \end{aligned}$$

   Defining $y \overset{\text{def}}{=} \log\{\Lambda(t)\}$ and $x \overset{\text{def}}{=} \log\{t\}$, this equation takes the linear form $y = a + bx$ with intercept $a = p\log\{\lambda\}$ and slope $b = p$.

□

[a] intro-to-survival-analysis.qmd#def-hazard
[b] https://morrison-lab.github.io/mds/algebra-exponentials.html#def-natural-log
[c] intro-to-survival-analysis.qmd#def-cuhaz

Example 1.3 (Evaluating Weibull hazard shapes under different shape parameters)**.** Consider a clinical trial evaluating time to relapse in months following cancer treatment, modeled using a Weibull distribution with scale parameter $\lambda = 0.1$. We evaluate the hazard rate $\lambda(t)$ at months $t = 1, 6, 12$ under three distinct shape parameters:

1. **Decreasing hazard ($p = 0.5$)**: The hazard rate formula gives:

$$\lambda(t) = (0.1)(0.5)(0.1t)^{-0.5} = \frac{0.05}{\sqrt{0.1t}}$$

   - At $t = 1$ month: $\lambda(1) = \frac{0.05}{\sqrt{0.1}} \approx 0.1581$ events per month.
   - At $t = 6$ months: $\lambda(6) = \frac{0.05}{\sqrt{0.6}} \approx 0.0645$ events per month.
   - At $t = 12$ months: $\lambda(12) = \frac{0.05}{\sqrt{1.2}} \approx 0.0456$ events per month. The risk of relapse is highest immediately following treatment and declines over time.
2. **Constant hazard ($p = 1.0$)**: The hazard rate formula reduces to $\lambda(t) = \lambda = 0.1000$ events per month for all t. The risk of relapse remains constant regardless of time elapsed.
3. **Increasing hazard ($p = 2.0$)**: The hazard rate formula gives:

$$\lambda(t) = (0.1)(2.0)(0.1t)^{1.0} = 0.02t$$

   - At $t = 1$ month: $\lambda(1) = 0.02(1) = 0.0200$ events per month.
   - At $t = 6$ months: $\lambda(6) = 0.02(6) = 0.1200$ events per month.
   - At $t = 12$ months: $\lambda(12) = 0.02(12) = 0.2400$ events per month. The risk of relapse increases linearly over time.

Exercise 1.1. Verify the calculations in Example 1.3 and explain how the shape parameter p affects cumulative hazard curves.

The Weibull distribution provides more flexibility than the exponential. Figure 2 shows some Weibull

hazard functions, with $\lambda = 1$ and p varying:

```
library(ggplot2)
library(eha)
lambda <- 1

ggplot() +
  geom_function(
    aes(col = "0.25"),
    fun = \(x) hweibull(x, shape = 0.25, scale = 1 / lambda)
  ) +
  geom_function(
    aes(col = "0.5"),
    fun = \(x) hweibull(x, shape = 0.5, scale = 1 / lambda)
  ) +
  geom_function(
    aes(col = "1"),
    fun = \(x) hweibull(x, shape = 1, scale = 1 / lambda)
  ) +
  geom_function(
    aes(col = "1.5"),
    fun = \(x) hweibull(x, shape = 1.5, scale = 1 / lambda)
  ) +
  geom_function(
    aes(col = "2"),
    fun = \(x) hweibull(x, shape = 2, scale = 1 / lambda)
  ) +
  theme_bw() +
  xlim(0, 2.5) +
  ylab(expr(lambda)) +
  theme(axis.title.y = element_text(angle = 0)) +
  theme(legend.position = "bottom") +
  guides(
    col =
      guide_legend(
        title = "p",
        label.theme = element_text(size = 12)
      )
  )
```

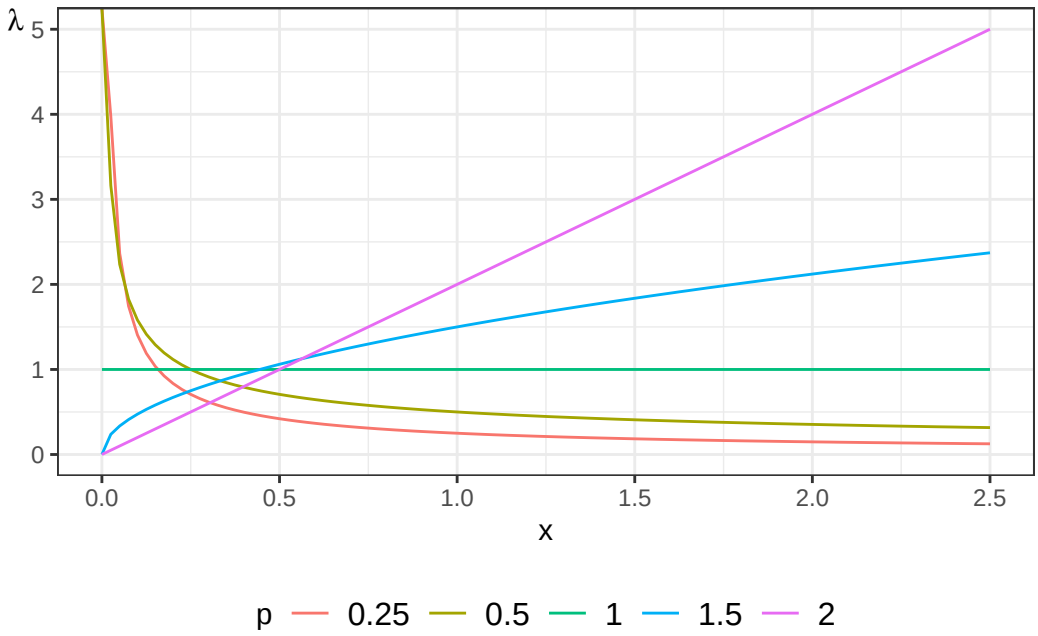


Figure 2: Hazard functions for Weibull distribution

```r
library(ggplot2)
lambda <- 1

ggplot() +
  geom_function(
    aes(col = "0.25"),
    fun = pweibull,
    args = list(shape = 0.25, scale = 1 / lambda, lower.tail = FALSE)
  ) +
  geom_function(
    aes(col = "0.5"),
    fun = pweibull,
    args = list(shape = 0.5, scale = 1 / lambda, lower.tail = FALSE)
  ) +
  geom_function(
    aes(col = "1"),
    fun = pweibull,
    args = list(shape = 1, scale = 1 / lambda, lower.tail = FALSE)
  ) +
  geom_function(
    aes(col = "1.5"),
    fun = pweibull,
    args = list(shape = 1.5, scale = 1 / lambda, lower.tail = FALSE)
  ) +
  geom_function(
    aes(col = "2"),
    fun = pweibull,
    args = list(shape = 2, scale = 1 / lambda, lower.tail = FALSE)
  ) +
  theme_bw() +
  xlim(0, 2.5) +
  ylab("Survival\nprobability,\nS(t)") +
```

```r
theme(axis.title.y = element_text(angle = 0)) +
theme(legend.position = "bottom") +
guides(
  col =
    guide_legend(
      title = "p",
      label.theme = element_text(size = 12)
    )
)
```

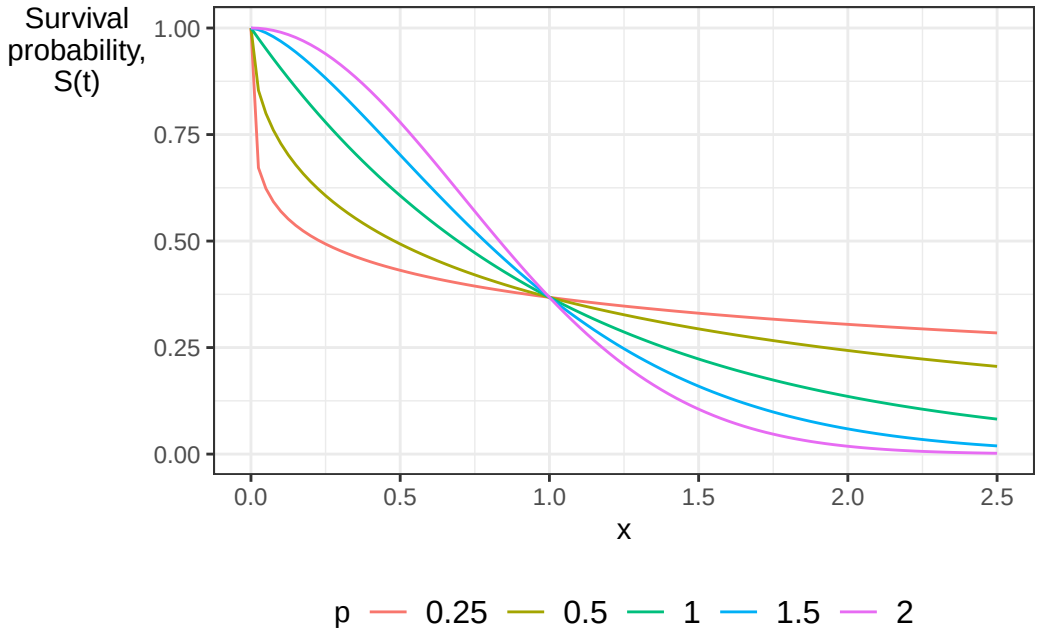


Figure 3: Survival functions for Weibull distribution

1.3 Exponential Regression

Definition 1.1 (Exponential regression model)**.** An **exponential regression model** for an event time T with predictor[a] vector $\underset{\sim}{x}$ assumes that, given $\underset{\sim}{x}$, T has an exponential distribution[b] whose hazard rate[c] is the exponentiated linear predictor[d]:

$$\lambda(t \mid \underset{\sim}{x}) \overset{\text{def}}{=} \exp\{\eta(\underset{\sim}{x})\}, \quad \eta(\underset{\sim}{x}) \overset{\text{def}}{=} \underset{\sim}{x} \cdot \underset{\sim}{\beta} = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p \tag{1}$$

[a] Intro-to-GLMs.qmd#def-predictors
[b] https://morrison-lab.github.io/pds/random-variables.html#def-exponential
[c] intro-to-survival-analysis.qmd#def-hazard
[d] Intro-to-GLMs.qmd#def-linear-predictor

Example 1.4 (Exponential regression with one binary covariate)**.** Let $x_1 = 1$ for patients on a standard treatment and $x_1 = 0$ for patients on a new treatment, and suppose $\beta_0 = \log\{0.025\}$ and $\beta_1 = \log\{4\}$, with time measured in weeks. By Equation 1:

$$
\begin{aligned}
\lambda(t \mid x_1 = 0) &= \exp\{\beta_0\} && \text{(exponential regression hazard with } x_1 = 0) \\
&= 0.025 && (\exp\{\log\{0.025\}\} = 0.025) \\
\lambda(t \mid x_1 = 1) &= \exp\{\beta_0 + \beta_1\} && \text{(exponential regression hazard with } x_1 = 1) \\
&= \exp\{\beta_0\} \cdot \exp\{\beta_1\} && \text{(product rule for exponentials)} \\
&= 0.025 \cdot 4 && \text{(evaluate each factor)} \\
&= 0.1 && \text{(multiply)}
\end{aligned}
$$

Each group's mean event time is the reciprocal of its hazard rate, as derived in Section 1.1: $1/0.025 = 40$ weeks on the new treatment and $1/0.1 = 10$ weeks on the standard treatment.

Exponential regression models the hazard rate using a log link function[5]. At $\underset{\sim}{x} = \underset{\sim}{0}$, Equation 1 gives the baseline hazard rate:

$$
\begin{aligned}
\lambda_0(t) &\overset{\text{def}}{=} \lambda(t \mid \underset{\sim}{0}) && \text{(baseline hazard rate definition at } \underset{\sim}{x} = \underset{\sim}{0}) \\
&= \exp\{\eta(\underset{\sim}{0})\} && \text{(substituting } \underset{\sim}{x} = \underset{\sim}{0} \text{ into hazard function)} \\
&= \exp\{\beta_0 + (\beta_1 \cdot 0 + \cdots + \beta_p \cdot 0)\} && \text{(evaluating linear predictor at zero vector)} \\
&= \exp\{\beta_0\} && \text{(simplifying exponent)}
\end{aligned}
$$

When there are no additional predictors[6] ($\underset{\sim}{x} = \underset{\sim}{0}$), the baseline hazard[7] is constant: $\lambda = \exp\{\beta_0\}$. This model uses a log link as in a generalized linear model[8] for Poisson[9] or rate outcomes. Because the hazard rate does not depend on time t, the hazard ratio[10] comparing two covariate vectors $\underset{\sim}{x}$ and $\underset{\sim}{x}^*$ is constant over time:

$$
\begin{aligned}
{}_\lambda(t \mid \underset{\sim}{x} : \underset{\sim}{x}^*) &\overset{\text{def}}{=} \frac{\lambda(t \mid \underset{\sim}{x})}{\lambda(t \mid \underset{\sim}{x}^*)} && \text{(hazard ratio definition)} \\
&= \frac{\exp\{\eta(\underset{\sim}{x})\}}{\exp\{\eta(\underset{\sim}{x}^*)\}} && \text{(substituting exponential regression hazard rates)} \\
&= \exp\{\eta(\underset{\sim}{x}) - \eta(\underset{\sim}{x}^*)\} && \text{(applying quotient rule for exponentials)} \\
&= \exp\{(\underset{\sim}{x} - \underset{\sim}{x}^*) \cdot \underset{\sim}{\beta}\} && \text{(expressing difference in linear predictors)}
\end{aligned}
$$

Thus, exponential regression models possess proportional hazards[11] trivially.

1.4 Accelerated Failure Time

In a proportional hazards (PH) model[12], covariates act multiplicatively on the baseline hazard function[13]:

[5] Intro-to-GLMs.qmd#def-link-function
[6] Intro-to-GLMs.qmd#def-predictors
[7] proportional-hazards-models.qmd#def-base-hazard
[8] Intro-to-GLMs.qmd#def-glm
[9] https://morrison-lab.github.io/pds/distributions.html#def-poisson
[10] intro-to-survival-analysis.qmd#def-hazard-ratio
[11] proportional-hazards-models.qmd#def-pha
[12] proportional-hazards-models.qmd#def-ph-model
[13] proportional-hazards-models.qmd#def-base-hazard

$$
\begin{aligned}
\lambda(t \mid \underset{\sim}{x}) &\stackrel{\text{def}}{=} \lambda_0(t) \cdot \theta_\lambda(\underset{\sim}{x}) && \text{(proportional hazards hazard rate definition)} \\
\Lambda(t \mid \underset{\sim}{x}) &\stackrel{\text{def}}{=} \int_0^t \lambda(u \mid \underset{\sim}{x}) du && \text{(cumulative hazard function definition)} \\
&= \int_0^t \lambda_0(u) \cdot \theta_\lambda(\underset{\sim}{x}) du && \text{(substituting proportional hazard rate)} \\
&= \theta_\lambda(\underset{\sim}{x}) \cdot \Lambda_0(t) && \text{(factoring constant hazard multiplier out of integral)} \\
\mathrm{S}(t \mid \underset{\sim}{x}) &\stackrel{\text{def}}{=} \exp\{-\Lambda(t \mid \underset{\sim}{x})\} && \text{(survival function in terms of cumulative hazard)} \\
&= \exp\{-\theta_\lambda(\underset{\sim}{x}) \cdot \Lambda_0(t)\} && \text{(substituting cumulative hazard)} \\
&= [\exp\{-\Lambda_0(t)\}]^{\theta_\lambda(\underset{\sim}{x})} && \text{(applying exponent laws)} \\
&= [\mathrm{S}_0(t)]^{\theta_\lambda(\underset{\sim}{x})} && \text{(substituting baseline survival function } \mathrm{S}_0(t)\text{)}
\end{aligned}
$$

An alternative modeling framework is the **Accelerated Failure Time (AFT)** model. Rather than multiplying the hazard rate, covariates accelerate or decelerate the progression of time itself.

Definition 1.2 (Acceleration factor)**.** For a covariate vector $\underset{\sim}{x}$ and coefficients $\beta_1, \ldots, \beta_p$, let $\eta(\underset{\sim}{x}) \stackrel{\text{def}}{=} \beta_1 x_1 + \cdots + \beta_p x_p$, with no intercept. The **acceleration factor** is

$$\theta_\lambda(\underset{\sim}{x}) \stackrel{\text{def}}{=} \exp\{\eta(\underset{\sim}{x})\}.$$

Example 1.5 (Computing an acceleration factor)**.** Suppose there is one covariate, with coefficient $\beta_1 = \log\{2\}$, and a subject has $x_1 = 2$. Then:

$$
\begin{aligned}
\theta_\lambda(\underset{\sim}{x}) &= \exp\{\eta(\underset{\sim}{x})\} && \text{(definition of the acceleration factor)} \\
&= \exp\{\beta_1 x_1\} && \text{(one covariate, no intercept)} \\
&= \exp\{2\log\{2\}\} && \text{(substitute } \beta_1 = \log\{2\} \text{ and } x_1 = 2\text{)} \\
&= 4 && \text{(}\exp\{\log\{a\}\} = a\text{, and } 2\log\{2\} = \log\{4\}\text{)}
\end{aligned}
$$

Definition 1.3 (Accelerated failure time model)**.** An **accelerated failure time (AFT) model** for an event time T with covariate vector $\underset{\sim}{x}$ assumes that

$$\mathrm{S}(t \mid \underset{\sim}{x}) \stackrel{\text{def}}{=} \mathrm{S}_0(t \cdot \theta_\lambda(\underset{\sim}{x})) \tag{2}$$

where:

- $\mathrm{S}_0(t) \stackrel{\text{def}}{=} \mathbb{P}(T > t \mid \underset{\sim}{x} = \underset{\sim}{0})$ is the baseline survival function[a], and
- $\theta_\lambda(\underset{\sim}{x})$ is the acceleration factor (Definition 1.2).

[a] proportional-hazards-models.qmd#def-baseline-surv

An acceleration factor $\theta_\lambda(\underset{\sim}{x}) > 1$ makes time pass faster for a subject with covariates $\underset{\sim}{x}$ than at baseline, so events happen sooner; $\theta_\lambda(\underset{\sim}{x}) < 1$ slows time down, so events happen later.

Example 1.6 (An acceleration factor of 4)**.** Suppose the baseline survival function is exponential with mean 40 weeks, $\mathrm{S}_0(t) = \exp\{-t/40\}$, and a covariate pattern $\underset{\sim}{x}$ has acceleration factor $\theta_\lambda(\underset{\sim}{x}) = 4$. By Equation 2, the probability that a subject with covariates $\underset{\sim}{x}$ is still event-free at 10 weeks is:

$$
\begin{aligned}
\mathrm{S}(10 \mid \underset{\sim}{x}) &= \mathrm{S}_0(10 \cdot 4) && \text{(AFT survival function with } t = 10 \text{ and } \theta_\lambda(\underset{\sim}{x}) = 4) \\
&= \mathrm{S}_0(40) && \text{(multiply)} \\
&= \exp\{-40/40\} && \text{(substitute the baseline survival function)} \\
&= \exp\{-1\} && \text{(divide)} \\
&\approx 0.368 && \text{(evaluate)}
\end{aligned}
$$

A subject with covariates $\underset{\sim}{x}$ has the same chance of lasting 10 weeks that a baseline subject has of lasting 40 weeks.

Theorem 1.3 (AFT models rescale every quantile)**.** *In an AFT model (Definition 1.3), suppose the baseline survival function* S_0 *is continuous and strictly decreasing, and let* t_q^0 *be the baseline time by which a fraction* q *of subjects have had the event, so that* $\mathrm{S}_0(t_q^0) = 1 - q$*. Then the corresponding time for covariate pattern* $\underset{\sim}{x}$ *is*

$$
t_q(\underset{\sim}{x}) = \frac{t_q^0}{\theta_\lambda(\underset{\sim}{x})}.
$$

Proof

Proof. The time $t_q(\underset{\sim}{x})$ solves $\mathrm{S}(t_q(\underset{\sim}{x}) \mid \underset{\sim}{x}) = 1 - q$:

$$
\begin{aligned}
1 - q &= \mathrm{S}(t_q(\underset{\sim}{x}) \mid \underset{\sim}{x}) && \text{(definition of } t_q(\underset{\sim}{x})) \\
&= \mathrm{S}_0\big(t_q(\underset{\sim}{x}) \cdot \theta_\lambda(\underset{\sim}{x})\big) && \text{(AFT survival function)}
\end{aligned}
$$

Because S_0 is strictly decreasing, t_q^0 is the only time at which S_0 equals $1 - q$, so:

$$
\begin{aligned}
t_q(\underset{\sim}{x}) \cdot \theta_\lambda(\underset{\sim}{x}) &= t_q^0 && (\mathrm{S}_0 \text{ is one-to-one}) \\
t_q(\underset{\sim}{x}) &= \frac{t_q^0}{\theta_\lambda(\underset{\sim}{x})} && (\text{divide by } \theta_\lambda(\underset{\sim}{x}) > 0)
\end{aligned}
$$

□

Example 1.7 (Median event times with an acceleration factor of 4)**.** In Example 1.6, the baseline survival function is $\mathrm{S}_0(t) = \exp\{-t/40\}$. Its median ($q = 0.5$) solves $\mathrm{S}_0(t_{0.5}^0) = 0.5$:

$$
\begin{aligned}
\exp\{-t_{0.5}^0/40\} &= 0.5 && \text{(definition of the median)} \\
-t_{0.5}^0/40 &= \log\{0.5\} && \text{(take logs)} \\
t_{0.5}^0 &= -40\log\{0.5\} && \text{(multiply by } -40) \\
&= 40\log\{2\} && (-\log\{0.5\} = \log\{1/0.5\} = \log\{2\}) \\
&\approx 27.73 && \text{(evaluate)}
\end{aligned}
$$

By Theorem 1.3, with $\theta_\lambda(\underset{\sim}{x}) = 4$, the median for covariate pattern $\underset{\sim}{x}$ is $27.73/4 \approx 6.93$ weeks. To check: $\mathrm{S}(6.93 \mid \underset{\sim}{x}) = \mathrm{S}_0(6.93 \cdot 4) = \mathrm{S}_0(27.73) = 0.5$.

We derive the expected survival time under the AFT model using change-of-variables integration:

$$
\begin{aligned}
\mathbb{E}[T \mid \underset{\sim}{x}] &\overset{\text{def}}{=} \int_0^\infty \mathrm{S}(t \mid \underset{\sim}{x}) dt && \text{(expectation formula for non-negative random variable)} \\
&= \int_0^\infty \mathrm{S}_0(t \cdot \theta_\lambda(\underset{\sim}{x})) dt && \text{(substituting AFT survival function)} \\
&= \int_0^\infty \mathrm{S}_0(u) \cdot \frac{du}{\theta_\lambda(\underset{\sim}{x})} && \text{(substituting } u = t \cdot \theta_\lambda(\underset{\sim}{x}) \implies dt = \frac{du}{\theta_\lambda(\underset{\sim}{x})}\text{)} \\
&= \theta_\lambda(\underset{\sim}{x})^{-1} \cdot \int_0^\infty \mathrm{S}_0(u) du && \text{(factoring constant } \theta_\lambda(\underset{\sim}{x})^{-1} \text{ out of integral)} \\
&= \theta_\lambda(\underset{\sim}{x})^{-1} \cdot \mathbb{E}[T \mid \underset{\sim}{x} = \underset{\sim}{0}] && \text{(substituting baseline expectation formula)}
\end{aligned}
$$

Thus, the mean survival time for a subject with covariates $\underset{\sim}{x}$ equals the baseline mean survival time divided by the acceleration factor $\theta_\lambda(\underset{\sim}{x}) = \exp\{\eta(\underset{\sim}{x})\}$.

We also derive the cumulative hazard and hazard rate functions under the AFT formulation:

$$
\begin{aligned}
\Lambda(t \mid \underset{\sim}{x}) &\overset{\text{def}}{=} -\log\{\mathrm{S}(t \mid \underset{\sim}{x})\} && \text{(cumulative hazard function definition)} \\
&= -\log\{\mathrm{S}_0(t \cdot \theta_\lambda(\underset{\sim}{x}))\} && \text{(substituting AFT survival function)} \\
&= \Lambda_0(t \cdot \theta_\lambda(\underset{\sim}{x})) && \text{(substituting baseline cumulative hazard)} \\
\lambda(t \mid \underset{\sim}{x}) &\overset{\text{def}}{=} \frac{\partial}{\partial t} \Lambda(t \mid \underset{\sim}{x}) && \text{(hazard function definition)} \\
&= \frac{\partial}{\partial t} (\Lambda_0(t \cdot \theta_\lambda(\underset{\sim}{x}))) && \text{(substituting AFT cumulative hazard)} \\
&= \Lambda_0'(t \cdot \theta_\lambda(\underset{\sim}{x})) \cdot \frac{\partial}{\partial t}(t \cdot \theta_\lambda(\underset{\sim}{x})) && \text{(applying chain rule of differentiation)} \\
&= \lambda_0(t \cdot \theta_\lambda(\underset{\sim}{x})) \cdot \theta_\lambda(\underset{\sim}{x}) && \text{(since } \Lambda_0'(u) = \lambda_0(u)\text{)} \\
&= \theta_\lambda(\underset{\sim}{x}) \cdot \lambda_0(t \cdot \theta_\lambda(\underset{\sim}{x})) && \text{(rearranging factors)}
\end{aligned}
$$

1.4.1 Relationship between PH and AFT models

If the baseline distribution is exponential with parameter λ, substituting $\mathrm{S}_0(t) = \exp\{-\lambda t\}$ into the AFT model yields:

$$
\begin{aligned}
\mathrm{S}(t \mid \underset{\sim}{x}) &= \exp\{-\lambda \cdot t \cdot \theta_\lambda(\underset{\sim}{x})\} && \text{(substituting exponential baseline into AFT formula)} \\
&= [\exp\{-\lambda t\}]^{\theta_\lambda(\underset{\sim}{x})} && \text{(applying exponent laws)} \\
&= [\mathrm{S}_0(t)]^{\theta_\lambda(\underset{\sim}{x})} && \text{(substituting exponential baseline survival function)}
\end{aligned}
$$

This derivation demonstrates that the exponential AFT model is mathematically identical to the exponential PH model.

The Weibull distribution has the same property.

Theorem 1.4 (A Weibull proportional hazards model is an AFT model)**.** *Let* $\mathrm{S}_0(t) = \exp\{-(\lambda t)^p\}$ *be a Weibull baseline survival function with scale* $\lambda > 0$ *and shape* $p > 0$*. If subject* i *has hazard* $\lambda_0(t) \cdot \theta_i$ *for a constant* $\theta_i > 0$*, then subject* i*'s survival function is*

$$\mathrm{S}_i(t) = \mathrm{S}_0\big(t \cdot \theta_i^{1/p}\big),$$

so the model is also an accelerated failure time model, with acceleration factor $\theta_i^{1/p}$*.*

Proof

Proof.

$$
\begin{aligned}
\mathrm{S}_i(t) &\overset{\text{def}}{=} \exp\left\{-\int_0^t \lambda_0(u)\cdot\theta_i\,du\right\} && \text{(survival function in terms of the hazard)}\\
&= \exp\{-\theta_i\cdot\Lambda_0(t)\} && \text{(factor the constant } \theta_i \text{ out of the integral)}\\
&= \exp\{-\theta_i(\lambda t)^p\} && \text{(Weibull cumulative hazard } \Lambda_0(t)=(\lambda t)^p)\\
&= \exp\left\{-\left(\theta_i^{1/p}\right)^p(\lambda t)^p\right\} && \text{(write } \theta_i=\left(\theta_i^{1/p}\right)^p)\\
&= \exp\left\{-\left(\lambda\cdot t\cdot\theta_i^{1/p}\right)^p\right\} && \text{(combine the } p\text{-th powers)}\\
&= \mathrm{S}_0\left(t\cdot\theta_i^{1/p}\right) && \text{(Weibull baseline survival function at } t\cdot\theta_i^{1/p})
\end{aligned}
$$

□

Example 1.8 (A hazard ratio of 4 with Weibull shape 2)**.** Take $\lambda = 0.05$, $p = 2$, and $\theta_i = 4$, with time in weeks. By Theorem 1.4, the acceleration factor is

$$
\begin{aligned}
\theta_i^{1/p} &= 4^{1/2} && \text{(substitute } \theta_i=4,\ p=2)\\
&= 2 && \text{(square root)}
\end{aligned}
$$

To check, compute subject i's survival at 10 weeks both ways:

$$
\begin{aligned}
[\mathrm{S}_0(10)]^{\theta_i} &= [\exp\{-(0.05\cdot 10)^2\}]^4 && \text{(proportional hazards form)}\\
&= \exp\{-4\cdot 0.25\} && ((0.5)^2=0.25\text{; exponent laws})\\
&= \exp\{-1\} && \text{(multiply)}\\
\mathrm{S}_0(10\cdot 2) &= \exp\{-(0.05\cdot 20)^2\} && \text{(AFT form with acceleration factor 2)}\\
&= \exp\{-1\} && ((1)^2=1)
\end{aligned}
$$

Both forms give $\exp\{-1\} \approx 0.368$: a four-fold hazard means that subject i ages twice as fast as baseline.

Theorem 1.5 (The Weibull model is a log-linear model for event times)**.** *Suppose subject i has the Weibull proportional hazards survival function $\mathrm{S}_i(t) = \exp\{-\theta_i(\lambda t)^p\}$ of Theorem 1.4, with $\theta_i \overset{def}{=} \exp\{\eta_i\}$ for a linear predictor η_i. Define $\alpha \overset{def}{=} -\log\{\lambda\}$ and $\sigma \overset{def}{=} 1/p$, and let W have the standard (minimum) Gumbel distribution, $\mathbb{P}(W > w) = \exp\{-e^w\}$. Then the log event time $Y_i \overset{def}{=} \log\{T_i\}$ has the same distribution as*

$$\alpha - \sigma\eta_i + \sigma W. \tag{3}$$

Proof

Proof. For any real y:

$$
\begin{aligned}
\mathbb{P}(Y_i > y) &= \mathbb{P}(T_i > \mathrm{e}^y) && (\log \text{ is increasing}) \\
&= \exp\{-\theta_i(\lambda \mathrm{e}^y)^p\} && (\text{Weibull PH survival function at } \mathrm{e}^y) \\
&= \exp\{-\mathrm{e}^{\eta_i}(\lambda \mathrm{e}^y)^p\} && (\theta_i = \mathrm{e}^{\eta_i}) \\
&= \exp\{-\mathrm{e}^{\eta_i}\lambda^p \mathrm{e}^{py}\} && (\text{power of a product}) \\
&= \exp\left\{-\mathrm{e}^{\eta_i}\mathrm{e}^{p\log\{\lambda\}}\mathrm{e}^{py}\right\} && (\lambda^p = \mathrm{e}^{p\log\{\lambda\}}) \\
&= \exp\left\{-\mathrm{e}^{\eta_i + p\log\{\lambda\} + py}\right\} && (\text{product rule for exponentials}) \\
&= \exp\left\{-\mathrm{e}^{\eta_i + (y-\alpha)/\sigma}\right\} && (\text{substitute } p = 1/\sigma,\ \log\{\lambda\} = -\alpha) \\
&= \exp\left\{-\mathrm{e}^{(y - \alpha + \sigma\eta_i)/\sigma}\right\} && (\text{common denominator } \sigma) \\
&= \mathbb{P}\left(W > \frac{y - \alpha + \sigma\eta_i}{\sigma}\right) && (\text{Gumbel survival function}) \\
&= \mathbb{P}(\sigma W > y - \alpha + \sigma\eta_i) && (\text{multiply both sides by } \sigma > 0) \\
&= \mathbb{P}(\alpha - \sigma\eta_i + \sigma W > y) && (\text{add } \alpha - \sigma\eta_i \text{ to both sides})
\end{aligned}
$$

The two random variables have the same survival function, so they have the same distribution. □

Example 1.9 (Median event times from the log-linear form)**.** Take $\lambda = 0.1$ and $p = 2$, so $\alpha = -\log\{0.1\} = \log\{10\} \approx 2.3026$ and $\sigma = 0.5$. The median of W solves $\exp\{-\mathrm{e}^w\} = 0.5$, so $\mathrm{e}^w = \log\{2\}$ and $w = \log\{\log\{2\}\} \approx -0.3665$. Because $\alpha - \sigma\eta + \sigma W$ is an increasing function of W, the median of Y is that function evaluated at the median of W.
For a baseline subject ($\eta = 0$):

$$
\begin{aligned}
\text{median}(Y) &= \alpha + \sigma \cdot (-0.3665) && (\text{log-linear form at the median of } W) \\
&= 2.3026 - 0.1833 && (\text{multiply}) \\
&= 2.1193 && (\text{subtract})
\end{aligned}
$$

so the median event time is $\mathrm{e}^{2.1193} \approx 8.33$. Directly, $\mathrm{S}_0(8.33) = \exp\{-(0.833)^2\} = \exp\{-0.694\} \approx 0.5$.
For a subject with hazard ratio $\theta = 4$, so $\eta = \log\{4\} \approx 1.3863$, the median of Y shifts by $-\sigma\eta = -0.5 \cdot 1.3863 \approx -0.6931$, so the median event time is multiplied by $\mathrm{e}^{-0.6931} = 0.5$, giving $8.33 \cdot 0.5 \approx 4.16$.

By Equation 3, the estimated AFT regression coefficients $\hat{\beta}_{\text{AFT}}$ produced by R's `survreg()` function relate directly to the PH regression coefficients $\hat{\beta}_{\text{PH}}$ produced by `coxph()` via the identity $\hat{\beta}_{\text{AFT}} = -\sigma\hat{\beta}_{\text{PH}}$.

The exponential and Weibull distributions are the only continuous distributions that are simultaneously proportional hazards models and accelerated failure time models. Other parametric families commonly used for survival data (log-logistic, log-normal, and generalized gamma) are AFT families rather than PH families; the log-logistic distribution is additionally a proportional-odds model.

1.5 Dataset: Leukemia treatments

To illustrate semi-parametric and parametric survival models, we analyze remission survival times from a clinical trial of 42 pediatric leukemia patients [2]. Half of the patients received a new therapy (6-mercaptopurine, 6-MP) and half received a standard control treatment.

The variables in this dataset include:

- `survt`: Remission duration in weeks (T).
- `status`: Relapse status (`relapse` = 1 or `censored` = 0).
- `rx`: Treatment assignment (`new` vs `standard`).
- `sex`: Patient sex (`female` vs `male`).
- `surv`: Survival outcome object created with `Surv(time = survt, event = (status == "relapse"))`.

```r
library(haven)
library(survival)
anderson <-
  paste0(
    "http://web1.sph.emory.edu/dkleinb/allDatasets",
    "/surv2datasets/anderson.dta"
  ) |>
  read_dta() |>
  mutate(
    status = status |>
      case_match(
        1 ~ "relapse",
        0 ~ "censored"
      ),
    sex = sex |>
      case_match(
        0 ~ "female",
        1 ~ "male"
      ),

    rx = rx |>
      case_match(
        0 ~ "new",
        1 ~ "standard"
      ),

    surv = Surv(time = survt, event = (status == "relapse"))
  )

print(anderson)
```

1.5.1 Cox semi-parametric model

We fit a Cox proportional hazards model[14] comparing standard therapy to new therapy:

```r
anderson_cox0 <- coxph(
  formula = surv ~ rx,
  data = anderson
)
summary(anderson_cox0)
#> Call:
#> coxph(formula = surv ~ rx, data = anderson)
#>
#>   n= 42, number of events= 30
#>
#>             coef exp(coef) se(coef)    z Pr(>|z|)
#> rxstandard 1.572     4.817    0.412 3.81  0.00014 ***
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#>
#>            exp(coef) exp(-coef) lower .95 upper .95
#> rxstandard      4.82      0.208      2.15      10.8
#>
#> Concordance= 0.69  (se = 0.041 )
#> Likelihood ratio test= 16.4  on 1 df,   p=5e-05
#> Wald test            = 14.5  on 1 df,   p=1e-04
```

[14] proportional-hazards-models.qmd#def-ph-model

```
#> Score (logrank) test = 17.2  on 1 df,   p=3e-05
```

The estimated log hazard ratio[15] $\hat{\beta}_{\text{PH}}$ for standard treatment relative to new treatment is positive, indicating that standard therapy is associated with a higher hazard of relapse (shorter remission duration).

1.5.2 Weibull parametric model

We next fit a parametric Weibull accelerated failure time model using `survreg()`:

```
anderson_weib <- survreg(
  formula = surv ~ rx,
  data = anderson,
  dist = "weibull"
)
summary(anderson_weib)
#>
#> Call:
#> survreg(formula = surv ~ rx, data = anderson, dist = "weibull")
#>              Value Std. Error     z       p
#> (Intercept)  3.516      0.252 13.96 < 2e-16
#> rxstandard  -1.267      0.311 -4.08 4.5e-05
#> Log(scale)  -0.312      0.147 -2.12   0.034
#>
#> Scale= 0.732
#>
#> Weibull distribution
#> Loglik(model)= -106.6   Loglik(intercept only)= -116.4
#>  Chisq= 19.65 on 1 degrees of freedom, p= 9.3e-06
#> Number of Newton-Raphson Iterations: 5
#> n= 42
```

In `survreg()`, regression parameters are reported on the log-time scale ($Y = \log T = \alpha + \underset{\sim}{\beta}_{\text{AFT}}\underset{\sim}{x} + \sigma W$). The estimated coefficient $\hat{\beta}_{\text{AFT}}$ for standard treatment is negative, indicating a reduction in log survival time (faster progression to relapse). The scale estimate $\hat{\sigma}$ corresponds to $1/p$, and satisfies $\hat{\beta}_{\text{AFT}} = -\hat{\sigma}\hat{\beta}_{\text{PH}}$.

1.5.3 Exponential parametric model

Setting the scale parameter to $\sigma = 1$ ($p = 1$) yields the exponential parametric model:

```
anderson_exp <- survreg(
  formula = surv ~ rx,
  data = anderson,
  dist = "exp"
)
summary(anderson_exp)
#>
#> Call:
#> survreg(formula = surv ~ rx, data = anderson, dist = "exp")
#>              Value Std. Error     z       p
#> (Intercept)  3.686      0.333 11.06 < 2e-16
#> rxstandard  -1.527      0.398 -3.83 0.00013
#>
#> Scale fixed at 1
#>
#> Exponential distribution
#> Loglik(model)= -108.5   Loglik(intercept only)= -116.8
```

[15] intro-to-survival-analysis.qmd#def-hazard-ratio

```
#>  Chisq= 16.49 on 1 degrees of freedom, p= 4.9e-05
#> Number of Newton-Raphson Iterations: 4
#> n= 42
```

In the exponential model, $\hat{\beta}_{\text{AFT}} = -\hat{\beta}_{\text{PH}}$, so the AFT coefficient is equal in magnitude and opposite in sign to the Cox proportional hazards coefficient.

Example 1.10 (Exponential rate estimates for the leukemia trial)**.** By the MLE of the exponential rate parameter[a], each group's estimated hazard rate is its number of relapses divided by its total follow-up time:

```
anderson |>
  summarize(
    .by = rx,
    relapses = sum(status == "relapse"),
    follow_up = sum(survt),
    rate = relapses / follow_up
  )
#> # A tibble: 2 x 4
#>   rx       relapses follow_up   rate
#>   <chr>       <int>     <dbl>  <dbl>
#> 1 new             9       359 0.0251
#> 2 standard       21       182 0.115
```

For the new treatment, 9 relapses over 359 weeks of follow-up give $\hat{\lambda}_{\text{new}} = 9/359 \approx 0.0251$ relapses per week; for the standard treatment, 21 relapses over 182 weeks give $\hat{\lambda}_{\text{standard}} = 21/182 \approx 0.1154$. The estimated log hazard ratio is then:

$$\begin{aligned}
\log\left\{\frac{\hat{\lambda}_{\text{standard}}}{\hat{\lambda}_{\text{new}}}\right\} &= \log\left\{\frac{21/182}{9/359}\right\} && \text{(substitute the rate estimates)} \\
&= \log\left\{\frac{21 \cdot 359}{182 \cdot 9}\right\} && \text{(divide fractions)} \\
&= \log\left\{\frac{7539}{1638}\right\} && \text{(multiply)} \\
&\approx 1.527 && \text{(evaluate)}
\end{aligned}$$

The `survreg()` fit's `rxstandard` coefficient, -1.527, is the negative of this log hazard ratio, as the exponential AFT model predicts, and its intercept, 3.686, is $\log\{359/9\}$, the log of the estimated mean remission time on the new treatment.

[a] intro-to-survival-analysis.qmd#thm-mle-exp

1.5.4 Diagnostic: complementary log-log survival plot

To check whether a Weibull proportional hazards assumption[16] is reasonable, we inspect the complementary log-log transformation of the Kaplan-Meier survival curves[17], $\log(-\log \hat{\text{S}}(t)) = p \log \lambda + p \log t$:

```
# plot the cumulative hazard H(t) = -log(S(t)) against time on log-spaced axes,
# so that the ticks are in weeks and in units of cumulative hazard,
# rather than in the logs of those quantities;
# log10 is not finite at 0, so leave out the points where H(t) = 0
# (before the first event) and where S(t) = 0 (so that H(t) is infinite):
survfit(
  formula = surv ~ rx,
  data = anderson
) |>
```

[16] proportional-hazards-models.qmd#def-pha
[17] intro-to-survival-analysis.qmd#def-KM-estimator

```r
broom::tidy() |>
dplyr::mutate(cumhaz = -log(estimate)) |>
dplyr::filter(time > 0, cumhaz > 0, is.finite(cumhaz)) |>
ggplot() +
aes(x = time, y = cumhaz, col = strata) +
geom_step() +
theme_bw() +
scale_x_continuous(
  trans = "log10",
  labels = scales::label_number(drop0trailing = TRUE, big.mark = ","),
  name = "Time (weeks, log scale)"
) +
scale_y_continuous(
  trans = "log10",
  labels = scales::label_number(drop0trailing = TRUE, big.mark = ","),
  name = "Cumulative hazard H(t) (log scale)"
) +
theme(legend.position = "bottom", legend.title = element_blank())
```

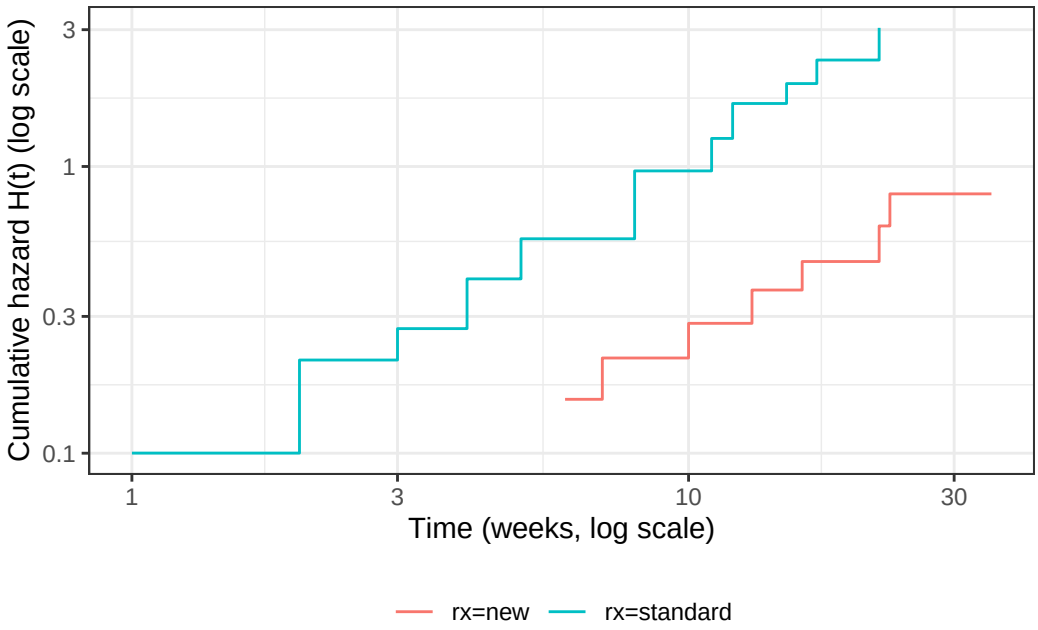


Figure 4: Cumulative hazard curves by treatment group, on log-spaced axes (complementary log-log survival curves)

If the complementary log-log curves are approximately linear and parallel across treatment groups, then a Weibull proportional hazards model is appropriate.

2 Combining left-truncation and interval-censoring

In observational studies and clinical trials, complex observation schemes can introduce both **left-truncation**[18] (delayed entry) and **interval-censoring**[19]. Left-truncation occurs when subjects enter the risk set[20] only after surviving past an entry time $L_{\text{entry}} > 0$. Interval-censoring occurs when the exact event time T_i is unknown, but is known to fall within an interval $(L_i, R_i]$.

[18] intro-to-survival-analysis.qmd#def-left-truncation
[19] intro-to-survival-analysis.qmd#def-interval-censoring
[20] intro-to-survival-analysis.qmd#def-risk-set

Standard software tools in R make distinct trade-offs between these two observation mechanisms:

> `coxph` does left truncation but not left (or interval) censoring; `survreg` does interval censoring but not left truncation (or time dependent covariates).
>
> — Terry Therneau, R-help, August 31, 2015[21]

In particular, `coxph()` constructs risk sets dynamically at each observed failure time t_j using counting process format (`time1 = entry`, `time2 = exit`), enabling seamless handling of left-truncation. However, because partial likelihood[22] relies on ordered exact failure times, standard Cox models cannot easily accommodate interval-censored data.

Conversely, `survreg()` accommodates interval-censored outcomes via `Surv(time1 = L, time2 = R, type = "interval")` by maximizing the interval parametric likelihood[23] $\mathbb{P}(L_i < T_i \leq R_i) = \mathrm{F}_0(R_i) - \mathrm{F}_0(L_i)$. However, standard `survreg()` assumes all subjects enter observation at time 0, and does not condition likelihood terms on surviving past a delayed entry time L_{entry}.

When a study contains both delayed entry and interval-censored event times, specialized tooling is needed. Among the following R packages, only `flexsurv` handles both features together; the other two are listed because they cover interval censoring well and are the usual starting points, not because they solve the truncation half:

- `flexsurv`: Fits flexible parametric survival models and allows custom likelihood specifications that incorporate both left-truncation conditioning and interval-censored bounds.
- `icenReg`: Fits regression models for interval-censored data: Cox proportional-hazards, proportional-odds, and accelerated failure time. Semi-parametric and fully parametric forms are both available, but the fully parametric option covers the AFT models only. It does not itself provide left-truncation conditioning.
- `interval`: Fits nonparametric survival curves (NPMLE) for interval-censored data and provides weighted logrank and Wilcoxon-type tests [3]. It is nonparametric throughout, so it supports neither regression nor truncation.

[1] D. M. Rocke, "Epi 204 lecture notes, Spring 2021." University of California, Davis, 2021. Available: https://dmrocke.ucdavis.edu/Class/EPI204-Spring-2021/EPI204-Spring-2021.html

[2] J. D. Kalbfleisch and R. L. Prentice, *The statistical analysis of failure time data*. John Wiley & Sons, 2011.

[3] M. P. Fay and P. A. Shaw, "Exact and asymptotic weighted logrank tests for interval censored data: The interval R package," *Journal of Statistical Software*, vol. 36, no. 2, pp. 1–34, 2010, doi: 10.18637/jss.v036.i02[24].

[21] https://stat.ethz.ch/pipermail/r-help/2015-August/431733.html
[22] proportional-hazards-models.qmd#def-ph-partial-lik
[23] https://morrison-lab.github.io/sds/intro-MLEs.html#def-lik
[24] https://doi.org/10.18637/jss.v036.i02